

MINI PROJECT

Mahindra Profit Analytics

Monthly Performance Report — Model-wise Sales, Margin & Profit Breakdown

Reporting period: Current month | Prepared for academic mini-project submission

₹3.57 Cr Total Monthly Profit	₹1.83 L Avg Profit / Car	190 Total Units Sold	XUV700 Top Profit Earner
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Executive Insight

Higher unit sales does not automatically translate into higher profit. The XUV300 sold the most units this month (50) but ranks only fourth in total profit (₹65 lakh), because its per-car margin (₹1.3 lakh) is the lowest of the five models. The XUV700, despite selling five fewer units, earns the most profit overall (₹1.125 crore) on the strength of a much higher per-car margin (₹2.5 lakh). Volume and profitability are two separate stories, and this report unpacks why.

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1. Introduction & Objective

This mini project analyses the monthly sales and profitability performance of five Mahindra SUV models — Bolero, Scorpio, Thar, XUV700 and XUV300 — sold through a dealership over one reporting month. The objective is to move beyond a simple sales-volume view and examine how much actual profit each model contributes, and why some high-selling models are not necessarily the most profitable.

Specifically, the project aims to:

- Compare unit sales against monthly profit for each model to expose any mismatch between volume and profitability.
- Quantify each model's per-car margin and its contribution to total monthly profit.
- Identify which models are volume leaders, margin leaders, and overall profit leaders.
- Translate the findings into actionable recommendations for a dealership's sales and inventory strategy.

2. Methodology & Data Sources

The analysis uses one month of dealership sales data covering units sold and profit per car for each of the five models. Monthly profit per model is derived as:

$$\text{Monthly Profit} = \text{Units Sold} \times \text{Profit per Car}$$

Profit share is each model's monthly profit expressed as a percentage of the total ₹3.57 crore pool. Charts were built to view the same underlying numbers from five different angles — ranked comparison, margin ranking, share of total, cumulative concentration, and a volume-vs-margin quadrant — so that both volume-driven and margin-driven stories are visible side by side.

3. Sales vs Profit — the Mismatch

Ranked by units sold, the chart below lines up sales volume against monthly profit for each model. Notice that the bars do not move together: the model with the tallest blue bar (most units sold, XUV300) is not the model with the tallest orange bar (most profit, XUV700).

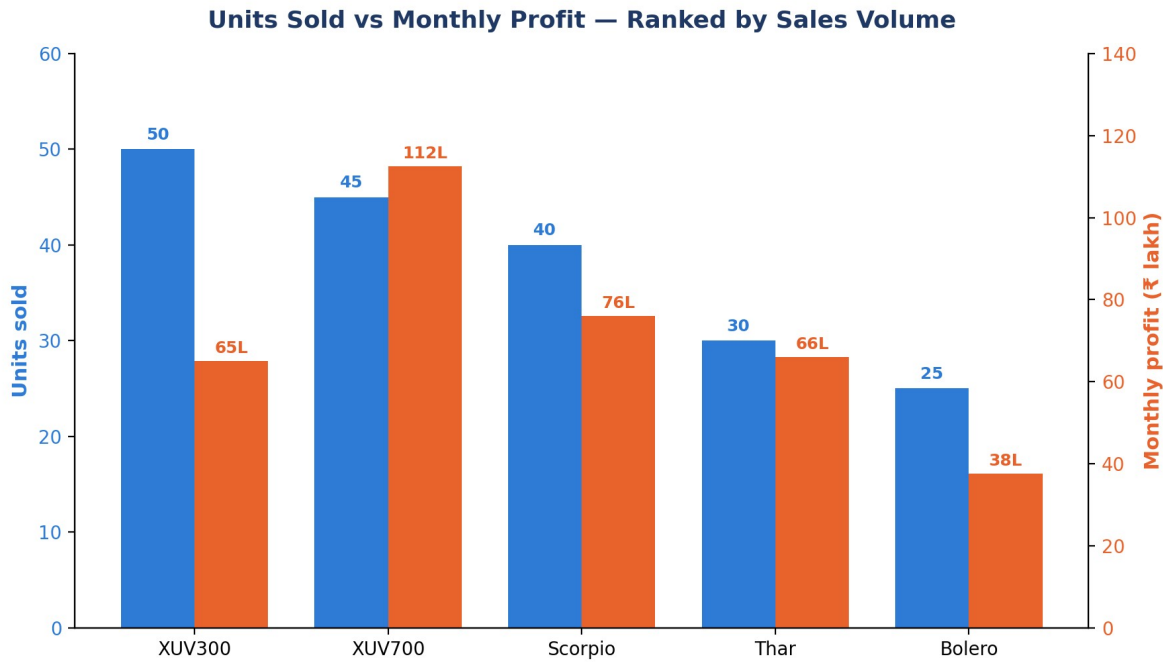


Figure 1: Units sold vs monthly profit, ranked by sales volume

XUV300 leads on units sold (50) but its profit bar (₹65L) is shorter than XUV700's (₹112.5L), even though XUV700 sold five fewer cars. This single chart is the clearest visual evidence that volume leadership and profit leadership are not the same thing in this dealership's line-up.

4. Profit Data Table

The table below is the source data behind every chart in this report, sorted alphabetically for reference. The amber row marks the highest-profit model (XUV700); the red row marks the highest-volume model that is not the highest-profit model (XUV300).

Model	Units Sold	Profit / Car (₹)	Monthly Profit (₹)	Share	Note
Bolero	25	1,50,000	37,50,000	10.5%	
Scorpio	40	1,90,000	76,00,000	21.3%	
Thar	30	2,20,000	66,00,000	18.5%	
XUV700	45	2,50,000	1,12,50,000	31.5%	Top profit
XUV300	50	1,30,000	65,00,000	18.2%	Top volume, weakest margin

Table 1: Model-wise units sold, per-car profit, monthly profit and profit share

Two numbers are worth comparing directly: XUV700's per-car margin (₹2,50,000) is roughly 1.9× XUV300's (₹1,30,000). That gap in margin is large enough to overturn a 5-unit sales deficit and still leave XUV700 nearly ₹47.5 lakh ahead in total profit.

5. Per-Unit Margin by Model

Viewed purely on per-car margin — independent of unit count — the ranking reshuffles completely from the sales-volume ranking. XUV700 sits clearly above the rest, followed by Thar, Scorpio, Bolero and XUV300.

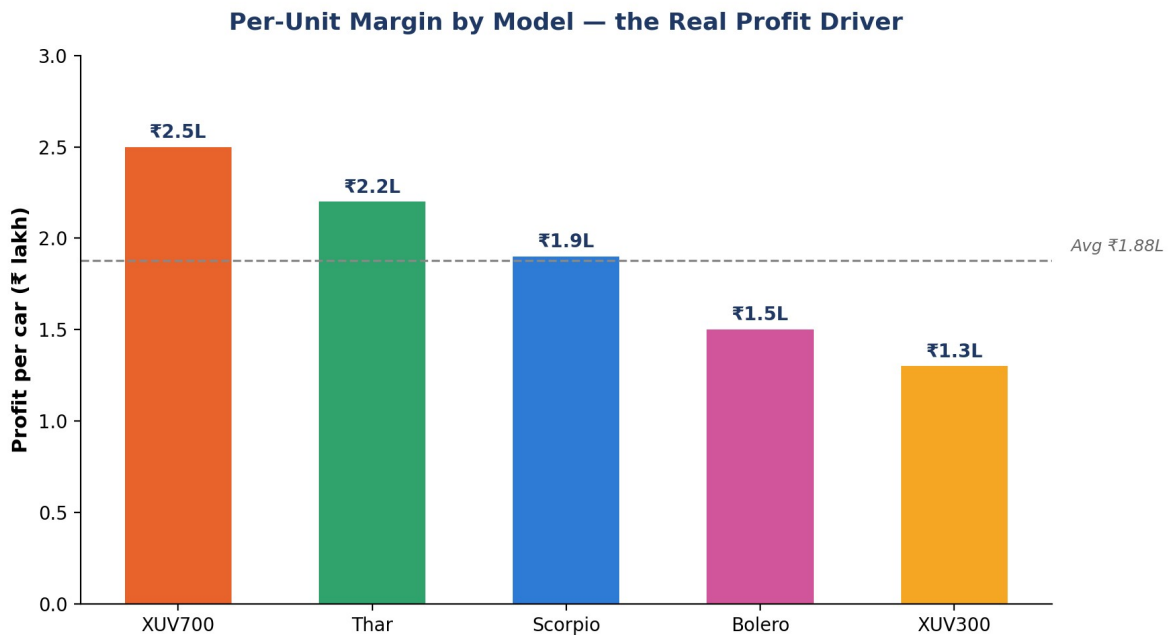


Figure 2: Per-unit margin by model (the underlying driver of Figure 1's mismatch)

This margin-only view explains the mismatch directly. XUV700 and Thar sit above the ₹1.88 lakh average per-car margin line, while XUV300 sits furthest below it. A model can be a volume leader and a margin laggard at the same time — that is exactly XUV300's position this month.

6. Profit Share Breakdown

Expressing each model's profit as a share of the ₹3.57 crore total shows how concentrated profitability is in the line-up.

Share of Total Monthly Profit by Model

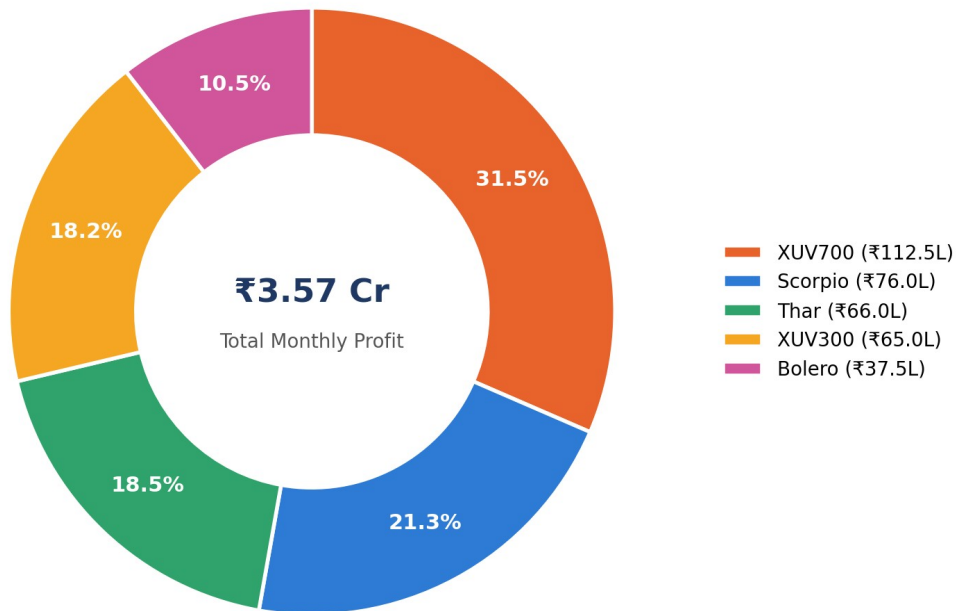


Figure 3: Share of total monthly profit by model

XUV700 alone accounts for 31.5% of total monthly profit — nearly one in every three rupees of profit earned this month came from a single model. Scorpio (21.3%) and Thar (18.5%) together add a further 39.8%, meaning three of the five models generate more than 70% of total profit between them, while Bolero contributes the smallest share at 10.5% despite not being the smallest-margin model.

7. Profit Concentration (Pareto Analysis)

Ranking models by profit and plotting cumulative contribution shows how quickly the total is reached as more models are added.

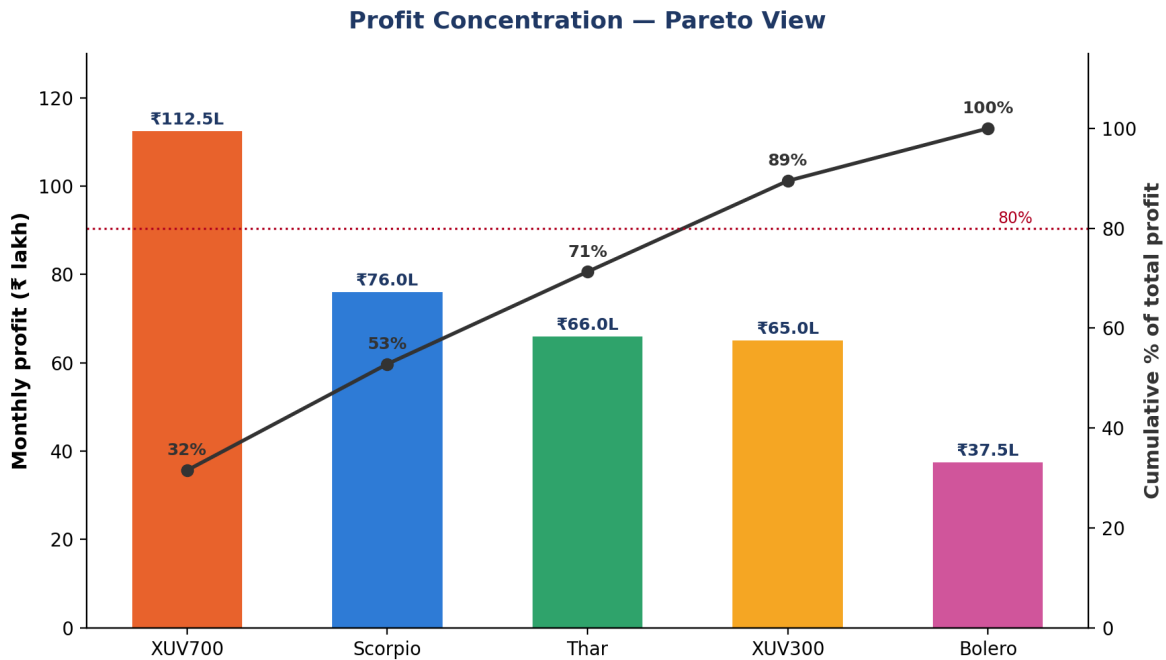


Figure 4: Cumulative profit contribution, ranked highest to lowest

The top two models by profit — XUV700 and Scorpio — already account for just under 53% of total monthly profit, and the top three (adding Thar) cross the 71% mark. The line does not reach the classic 80/20 extreme, but it confirms that profit is meaningfully front-loaded onto two or three models rather than spread evenly across all five.

8. Volume vs Margin Quadrant Analysis

Plotting units sold against profit per car — with bubble size representing total monthly profit — separates the five models into distinct strategic positions.

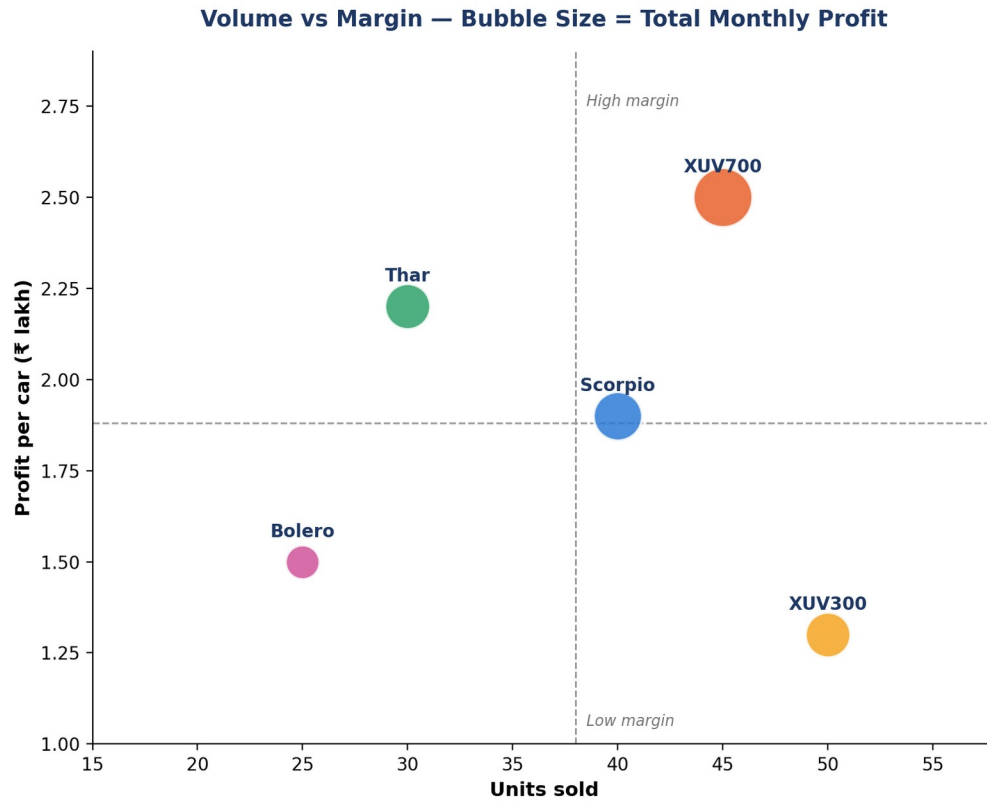


Figure 5: Volume vs margin, bubble size = total monthly profit

XUV700 sits in the high-margin, above-average-volume quadrant with the largest bubble — the strongest position in the line-up. XUV300 sits in the high-volume, low-margin quadrant: it moves the most metal but each sale contributes the least. Bolero is low-volume and mid-to-low margin, explaining its small bubble. Scorpio and Thar cluster near the centre, each balancing volume and margin reasonably evenly, which is why both land in the middle of the total-profit ranking.

9. Profit Flow to Total

Every model's monthly profit flows into the same ₹3.57 crore pool. The relative thickness of each contribution mirrors the profit share seen in Figure 3: Bolero, Scorpio, Thar and XUV300 each add a moderate stream, while XUV700 (highlighted) contributes the single largest stream into the total.

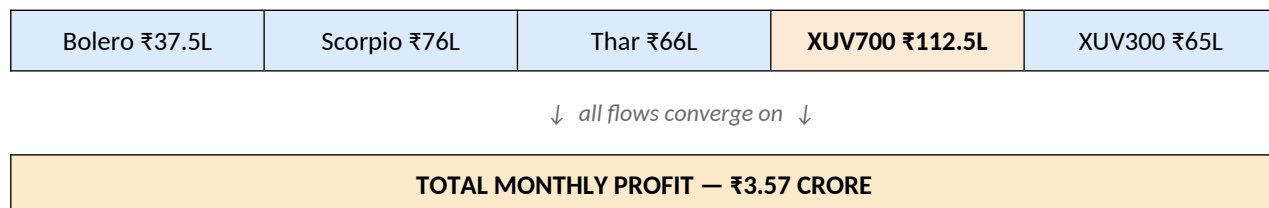


Figure 6: Simplified profit-flow summary

10. Key Takeaways

- **High sales does not equal high profit.** XUV300 sold the most units (50) but earned less total profit than Scorpio, Thar or XUV700 because its per-car margin is the lowest in the line-up.
- **XUV700 is the clear profit leader**, generating ₹1.125 crore per month (31.5% of total) on a strong per-car margin, despite selling fewer units than XUV300.
- **Scorpio delivers strong overall profit** (₹76 lakh, 21.3% share) from a healthy balance of volume and per-unit margin — it is the most 'balanced' performer.
- **Bolero sells the fewest units but still clears ₹37.5 lakh**, showing that a solid margin can partly offset lower volume.
- **Profit is concentrated, not even.** The top three models by profit (XUV700, Scorpio, Thar) generate over 71% of total monthly profit between them.
- Combined, the five models generate ₹3.57 crore in monthly profit at an average of ₹1.83 lakh profit per car sold.

11. Recommendations

Based on the volume-margin split observed above, the following actions follow directly from the data:

- **Push XUV700 availability and financing offers.** It is both a strong seller and the top margin earner — the safest model to prioritise in stock allocation and showroom floor space.
- **Investigate XUV300 pricing and cost structure.** Its margin is the weakest despite leading on volume; even a modest per-car margin improvement (via variant mix, accessory attach rate, or discount discipline) would have an outsized effect given its unit count.
- **Protect Scorpio and Thar's balanced position.** Both sit near the average on volume and margin — they are dependable profit contributors and good candidates for stable, repeatable sales targets.
- **Use Bolero's margin, not its volume, as the sales pitch.** With the lowest unit count, growth in Bolero profit is more likely to come from selling a few more units than from further margin expansion.
- **Track concentration risk.** With one model supplying nearly a third of total profit, a demand or supply shock to XUV700 (e.g. a stock shortage) would have a disproportionate effect on total monthly profit — worth monitoring alongside sales figures.

12. Limitations & Future Scope

- **Single-month snapshot:** the analysis covers one reporting month only; trends over multiple months would show whether this pattern is consistent or seasonal.
- **No cost breakdown:** profit per car is treated as a given figure; a future version could decompose it into selling price, dealer cost, discounts and incentives.
- **No regional or salesperson-level detail:** all figures are dealership-wide; splitting by location or sales team could reveal further variance.
- **Future scope:** extend to a rolling 12-month dashboard, add year-over-year comparisons, and correlate profit with marketing spend per model.

13. Conclusion

This mini project set out to test a simple but important business assumption — that higher sales volume drives higher profit — and found it does not hold uniformly across Mahindra's model line-up. XUV700 outperforms the volume leader, XUV300, on total profit purely through superior per-unit margin. For a dealership managing inventory, floor space, and sales incentives, this reframes the priority question from 'which model sells the most' to 'which model earns the most per unit sold' — and the data suggests these are two different questions with two different answers.

14. Model Reference Links

Official model pages, included for reference (see the original source PDF for live hyperlinks):

- Bolero — official page
- Scorpio — official page
- Thar — official page
- XUV700 — official page
- XUV300 — official page